

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

COMBINED SCIENCE
 PAPER 3 Practical Test

4003/3
 1 hour 30 minutes

SPECIMEN PAPER

Candidates answer on the question paper.

Additional materials:
 As listed in instructions to Supervisors
 Calculator (optional)
 30.0 cm ruler

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **both** questions.

Write your answers in the spaces provided on the question paper.

Use a sharp pencil for your drawings. Coloured pencils or crayons should **not** be used.

You should show the essential steps in any calculation and record all experimental results in the spaces provided in the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
TOTAL	

This specimen paper consists of 8 printed pages.

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1. You are required to determine the concentration of dilute sulphuric acid, H_2SO_4 through titration with sodium hydroxide solution (NaOH).

You are provided with dilute sulphuric acid, 1.0 mol/dm^3 sodium hydroxide solution, phenolphthalein indicator, a conical flask, a 25 cm^3 pipette, a pipette filler, a burette, a filter funnel, a dropper, a retort stand and clamp and a white tile.

Measure 25.00 cm^3 of the sodium hydroxide solution using the pipette.

Put the 25.00 cm^3 of the sodium hydroxide solution, NaOH , into the conical flask.

Using a dropper, add two or three drops of phenolphthalein indicator into the conical flask and shake gently.

Pour the dilute sulphuric acid into the burette (already mounted).

Read the initial volume of the dilute sulphuric acid.

Record the initial volume of the dilute sulphuric acid in **Table 1.1**.

Titrate the sodium hydroxide solution against the dilute sulphuric acid until the indicator just changes.

NB: Shake the conical flask continuously during the titration.

Record the final volume of the dilute sulphuric acid in the burette in **Table 1.1**.

Complete **Table 1.1** by calculating the volume of the dilute sulphuric acid, H_2SO_4 , used.

(a) (i) **Table 1.1**

experiment	rough	1	2
final burette reading/ cm^3			
initial burette reading/ cm^3			
volume of the H_2SO_4 used/ cm^3			

[9]

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- (ii) Calculate the average volume of the dilute sulphuric acid needed to react with 25.00 cm³ of sodium hydroxide solution, NaOH.

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[1]

- (iii) Explain why an indicator is used in the experiment.

_____ [2]

- (iv) State the type of reaction that occurs in the conical flask.

_____ [1]

- (v) Calculate the number of moles of sodium hydroxide in the 25.00 cm³ of solution.

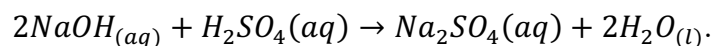
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[3]

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- (vi) Sodium hydroxide solution, NaOH, and dilute sulphuric acid, H₂SO₄ react according to the equation:



Determine the number of moles of the dilute sulphuric acid, H₂SO₄ which reacted with the 25.00 cm³ of sodium hydroxide solution, NaOH.

[1]

- (vii) Calculate the concentration of the dilute sulphuric acid, H₂SO₄.

[2]

- (b) Suggest **one** application of the titration procedure.

[1]

2. You are required to determine the effect of diameter on the resistance of a wire. You are provided with four wires of different diameters labelled **E**, **F**, **G** and **H**.

NB: The wires are of the same material and same length.

The circuit shown in **Fig 2.1** has been set up for you.

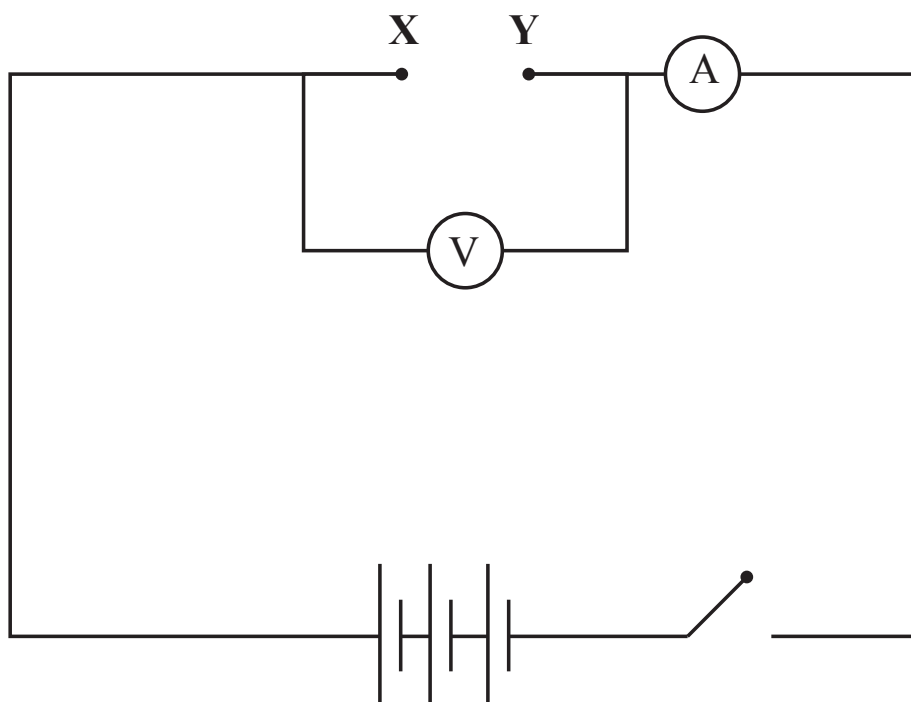


Fig 2.1

- (a) Connect wire **E** between point **X** and point **Y** using crocodile clips. Close the switch and record the ammeter and voltmeter readings in **Table 2.1**. Open the switch and remove wire **E**. Repeat the procedure using wires **F**, **G** and **H**.

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Complete **Table 2.1** by calculating the resistance.

Table 2.1

wire	diameter / mm	voltage / V	current / A	resistance / Ω
E	0.56			
F	0.46			
G	0.38			
H	0.25			

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- (b) (i) Plot a graph of resistance against diameter of the wires.



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(ii) Determine, clearly showing on the graph, the resistance when the diameter is 0.32 mm. [2]

(iii) Deduce a conclusion for the experiment.

_____ [1]

(c) Calculate the power of the circuit which uses wire G.

[3]

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